UNIVERSITI TUNKU ABDUL RAHMAN

ACADEMIC YEAR 2025/2026

OCTOBER 2025 EXAMINATION

UECS3253 WIRELESS APPLICATION DEVELOPMENT

MONDAY, 13 OCTOBER 2025 TIME : 9.00 AM – 11.00 AM (2 HOURS)

BACHELOR OF SOFTWARE ENGINEERING (HONOURS)

Instructions to Candidates:

This question paper consists of **TWO (2)** questions in **Section A** and **TWO (2)** questions in **Section B**.

Section A: Answer **ALL TWO (2)** questions.

Section B: Answer **ONLY ONE (1)** question.

Each question in **Section A** carries 30 marks, question in **Section B** carries 40 marks.

Remark: Should a candidate answer **TWO (2)** questions in **Section B**, marks will only be awarded for the **FIRST** question in the order the candidate submits the answers.

Answer the questions in the answer booklet provided.

UECS3253 WIRELESS APPLICATION DEVELOPMENT

Section A: Answer **ALL** questions.

Q1. (a) There are different types of mobile apps depending on the nature of the apps, including native apps, mobile web apps and hybrid apps.

(i) Identify **ONE (1)** platform-specific programming language to develop apps for iOS device. (2 marks)

(ii) Identify the name of app store that the developer can upload Android apps. (2 marks)

(iii) Identify **TWO (2)** different features of Android. (4 marks)

(b) Android is an open-source mobile operating system based on modified version of Linux.

(i) Android is based on mobile software stack that consists of different layers, identify **TWO (2)** layers in the mobile software stack. (4 marks)

(ii) Java API framework enables us to access the feature-set of the Android OS. Identify **ONE (1)** system component or service that could be used by the Java API framework. (2 marks)

(iii) React Native can be used to develop Android application. By referring to the command below, identify the suitable approach that the command is used to create the React Native project folder.

```
npx @react-native-community/cli init WAD --
version="0.80"
```

(2 marks)

(c) One of the important concepts of React Native is component.

(i) Consider the following code segment that defines a React Native component. Identify the suitable code to export ComponentA through default export.

```
export const ComponentA = () = {
    return(
        <>
        </>
    )
}
```

(2 marks)

Section A Q1. (c) (Continued)

(ii) Consider the following code segment that imports the React Native component. Identify the name of this import.

```
import * as module1 from './modules/module1';
```

(2 marks)

(iii) Lifecycle methods of React Native class components usually start with prefixes will or did. Explain their difference. (4 marks)

(iv) Given the code segment that implements class component, identify the functional component-based implementation for lines 8 - 10.

```
1:  // Assume the necessary Components were
2:  imported
3:
4:  export default class AppABC extends
5:  Component {
6:      constructor(props){
7:          super(props);
8:          this.state = {
9:            a: '0'
10:         }
11:     }
12:
13:    render() {
14:       return(
15:         <CustomComponent
16:           displayId={this.state.a}
17:         />
18:       );
19:    }
20: }
```

(2 marks)

(v) The code segment below has **TWO (2)** syntax errors, identify the correct implementation for the errors.

```
1:  // Assume the necessary Components were
2:  imported
3:
4:  export default function AppABC = () => {
5:    const [a, setA] = useState('0');
6:    return(
7:      <CustomComponent
8:        displayId={this.state.a}
9:      />
10:   );
11: }
```

(4 marks)
[Total : 30 marks]

Section A (Continued)

Q2. (a) Persistent data storage is essential as it retains user preferences or settings, enabling custom content to be displayed to users. The react-native-sqlite-storage library supports for using SQLite databases in React Native applications.

(i) Identify **ONE (1)** method that the JavaScript-based framework can access react-native-sqlite-storage. (2 marks)

(ii) By referring to the code segment below, identify when the methodA and methodB will be executed.

```
let db = SQLite.openDatabase({name:
'db.sqlite'}, methodA, methodB,
```

(4 marks)

(iii) Consider the developer is using React Native to develop a mobile LMS application. By referring to the code segment below, construct the complete code that to perform SQL query that adds the course in the table below to the database table courseInfo.

Table 2.1: courseInfo

courseId	courseCode	courseName
1	UECS3253	WAD

```
const methodQuestionTwo = async (
   db: SQLiteDatabase,
) => {

try {
    // Complete the code here

  } catch (error) {
    console.log(
  }
}
```

(4 marks)

(b) Other than SQLite, AsynStorage can be used to store data. AsyncStorage stores the data as key-value pair.

(i) Give the name of the library for React Native to use AsyncStorage. (2 marks)

(ii) By referring to the code segment below, construct the code to update the information that was stored using AsyncStorage. Update the value to 'abc' for each key in the courseInfoKey, and the function multiset should not be used.

Section A Q2. (b) (ii) (Continued)

```
// Assume the code is in an async function

const courseInfoKey = ['code', 'name'];

try {
    for (const key of courseInfoKey) {
        // Complete the code here
    }
} catch (error) {
    console.log(error);
}
```

(4 marks)

(iii) Considering the React Native application needs to store the password. Identify whether AsyncStorage is the suitable method for storing password and provide **ONE (1)** justification to support your answer.

(4 marks)

(c) React Native application can access the GPS module to obtain information regarding the location.

(i) Give the name of the file that the developer needs to modify so that the Android applications can request permission for location. (2 marks)

(ii) By referring to the code segment below, construct the complete code segment that allows the function getLink() to obtain the driving directions from Pasar Seni to Kepong. The information is obtained through Google service provider.

```
// Assume the necessary Components were imported

const data = {
    // Complete the code here
};

const getLink = () => {
     return createMapLink(data);
}
```

(8 marks)

[Total : 30 marks]

Section B: Answer ONLY **ONE (1)** question.

Q3. React Native applications can access different features by connecting to the servers.

(a) Consider the scenario that your company is developing a weather forecast application that connects to an API to obtain weather information. The connection is made using a RESTful API.

(i) Define web service and identify **TWO (2)** communication methods other than REST that the web service can use. (5 marks)

(ii) Provide **TWO (2)** similarities between APIs and web services. (4 marks)

(iii) Consider the server responds to the user with **THREE (3)** possible messages: (i) Moved Permanently, (ii) Unprocessable Entry and (iii) HTTP Version Not Supported. What are the corresponding HTTP status codes for each of these messages? (6 marks)

(b) The development team has implemented the Book Delete functionality using the Axios in the React Native application. Briefly, there were issues in the code which resulting the function not working properly. In this case, the development team has decided to switch to Fetch API.

```
// Assume the necessary Components were imported

const deleteBook = async (book) => {

    let url = "http://bookapp.com/api/deleteBook";

    try {
        // Re-implement this part
        const response = await axios.delete(url, {
            bookId: "1",
        });

        if (response.success) {
            ToastAndroid.display("Updated",
ToastAndroid.SHORT);
            navigation.go("BookInfo", { message:
"done"});
        } else {
            navigation.back();
        }
    } catch (error) {
        Alert.alert("Network Error");
    }
};
```

Section B Q3. (b) (Continued)

Construct the code snippet to replace the existing implementation from Axios to Fetch, while ensuring the following requirements are met:

Requirements:

- The application should send an **HTTP DELETE** request to delete the book at: **http://bookapp.com/api/deleteBook**.
- The data should be formatted as **JSON** with the following fields:
  - bookId: book.toString()
- If the deletion is **successful**, the application should show a **Toast message** and **navigate** to the "BookInfo" screen with passing the message "done" as a route parameter.
- If the update **fails**, the application should navigate back to the previous screen. (15 marks)

(c) Currently, you are required to develop the feature that allows the React Native application to retrieve the book information through the API endpoint that is hosted at http://bookapp.com/api/getBook.

```
// Assume the necessary Components were imported

const [book, setBook] = useState({});

const retrieveBook = async() => {

try {
        // Complete the code here
    } catch (error) {
        Alert.alert("Network Error");
    }
};
```

By referring to the code segment above, construct the complete code that uses Fetch API to connect to the server. Below are the requirements for the code:

- Use **HTTP GET** as the HTTP method.
- Convert the response to **JSON**.
- If successful, show a **Toast message** and set the value to employee.
- If HTTP status "OK" is received, show an **Alert message** and set the value to the state variable book.
- If HTTP status "OK" is not received, display a **Toast message** and navigate to the screen "mainScreen". (10 marks)

[Total : 40 marks]

Section B (Continued)

Q4. WebSocket allows persistent connection between client (React Native application) and server.

(a) Answer the following questions related to Socket.IO.

(i) By referring to the code segment below, identify the correction configuration so that the Socket.IO uses the mechanism where a client requests resource from the server and waits for the server. (1 mark)

```
var socket =
io('http://FE/api/sendEmail', {
    // Complete the code here
});
```

(ii) Identify the command to install the module that allows us to use Socket.IO in a React Native application. (1 mark)

(iii) Identify **TWO (2)** use cases for websockets. (4 marks)

(iv) There are a few functions related to the Web Socket in the React Native application, explain the difference between io(...) and emit(...). (4 marks)

(v) Considering there are **TWO (2)** URLs: (i) https://fe.com and (ii) https://fe.com/detection that are connected using function io(...) without additional options, identify whether the socket instances created in this case are using the same manager instance. (2 marks)

(vi) Identify **TWO (2)** IO factory options that can be specified when we create the socket instance through the function io(...). (4 marks)

(b) Consider the company is developing a React Native application that uses a traffic API to monitor traffic and track accidents that have occurred. Specifically, the server hosting the traffic API can be connected through Socket.IO. Complete the code segment below based on the requirement in the **Q4 (b) (i – iv)**, so that the React Native application can communicate with the traffic API.

(i) Construct code segment to create a web socket instance namely socketFE is created, and it is connected to the URL http://www.FE.com/trafficAPI. WebSocket and WebTransport are used as the communication methods. (6 marks)

(ii) Upon connected to the server, the React Native application displays an alert message "connected!" and set value of the trafficLocation to "Kajang". (6 marks)

Section B Q4. (b) (Continued)

(iii) Upon receiving the event "traffic_info", the handler first converts the data from JSON string to JSON object. After that, the React Native application displays the data through alert message and navigates back to the previous screen. (6 marks)

(iv) The React Native application submit trafficLocation data and a question "How is the traffic today" to the server. (6 marks)

```
// Assume the necessary components/functions
were imported

/*
  Complete the code here for Q4 (b) (i)
*/

const App = ({route, navigation}: any) => {

const [trafficLocation, setTrafficLocation] =
useState('');

useEffect(()=>{
    // A new handler is registered
    /*
        Complete the code here for Q4 (b) (ii)
    */


    // A new handler is registered
    /*
        Complete the code here for Q4 (b) (iii)
    */
},[]);

return(
    <ScrollView>

   <TextInput
        value={trafficLocation}
        onChangeText={(input) => {
          setTrafficLocation(input);
    }}
        placeholder="Enter location"
  />

  <Button
    title="Check Traffic"
    onPress={() => {
       /*
```

Section B Q4. (b) (iv) (Continued)

```
          Complete the code here for Q4 (b) (iv)
         */
      }}
   />     </ScrollView>
);
}
export default App;
```

[Total : 40 marks]